21 Feb 25 DASH Ask Me Event Questions

1. Will these slides be distributed?
    a. The slides have been posted as an RFI update in https://sam.gov/opp/041ba08481c14cd49a4644e734c58934/view.
    b. The recording of the AMA event is available upon request; please contact Lt Col Wintch if interested.
2. Will the scenario be available in advance so we can configure our system?
    a. Scenario specifics will not be available in advance. The situation will be an all-domain battle with dynamic targets.
3. Are there updates to the PAE in the model since last year? How will the Model be presented? Is it .MDZIP or something else?
    a. Yes. We will post the PAE model files as an RFI update to sam.gov and will be provided in MDZIP, HTML, and PDF formats.
4. Are there any updates to the TM beyond what have been published in SAM?
    a. Yes. The model for PAE will be posted.
5. Will the Version 3 TM be published ahead of the DASH, or will teams first receive it as the DASH?
    a. The Version 3 of the TM is still ongoing for function decomposition at the Tier 3 Level. We will post the PAE models ahead of the DASH.
6. Could you post a diagram, details, etc. that outlines the dev environment?
    a. Yes. Once the experiment plan has matured, we'll post an update to the RFI in sam.gov
7. Could example output formats be provided ahead of participation?
    a. To prevent constraining the development of any kind, the creative problem solving, possibility of many solutions, there will be no restrictions on output formats.
8. Will the test data be CUI or unclassified?
    a. The test data will be unclassified only.
9. Will the government provide any funding to offset the cost to industry for participating in this event?
    a. The Government is working on setting up a vehicle to support this, but there will be no funding available for the event beginning 31 March.
10. Will the development teams need to entirely be onsite or is remote access available?
    a. Remote access is available. The H2O Center has commercial (non-DoD NIPRnet) wireless internet available for industry to link back to their remote locations. Development teams need to have the right personnel on site to engage with the battle managers and the scenario to conduct work on solution development. Lack of on-site personnel could hinder real time engagement with the scenario.

11. Are we able to bring existing software along with us? Are we able to access this software via internet, or should we bring it on a laptop?
    a. Yes, if you can bring whatever solutions (your own software, laptops, etc.) you have, to support this experiment. The H2O will provide commercial wifi access.
12. Just to clarify, is there any restriction that would prevent vendors from using the wifi to reach cloud-based resources?
    a. No. There are no restrictions on the H2O commercially provided wi-fi internet.
13. What output format(s) are expected from the PAE microservices? Human readable prompts, M2M (e.g. UCI)?
    a. At the very least, it needs to be human consumable, interactable, and exploitable without being complicated. Simplicity is best. Battle Performers need to be able to understand the data and act on it.
14. Have you determined MOE/MOPs? How will we be graded?
    a. The model provides those MoEs and MoPs and you can see some of the more top-level measurements that we will use to assess. Since this an experiment event, there is no grading criteria to the solution development. The intent is to be able to develop some form of something that aides in Human-Machine Teaming capability.
15. Will you give us the types and scale of the context documents and the "deluge of inputs" to help us prepare our compute requirements?
    a. Yes. You will receive a text document, ~10 pages, which will contain the scenario context, essential elements of a plan, etc.
16. How many companies will be invited?
    a. Five companies will be selected for participation.
17. Is the dash event at the end of March/early April 2025 when performers would demonstrate their capabilities, or would that happen later in the program?
    a. The demonstration will happen as part of the March/April experiment event.
18. Is the recommendation from the microservice supposed to just contain recommended actions or the justifications/rationale as well for the recommendation? And is this also providing a decision matrix based on evaluation criteria?
    a. The point of these microservices is to be a decision-making teammate with a human operator. If I was teaming with another human, they might give me a quick answer and I might use it right away. But if there's time, I might want to ask them, hey, what was the rationale? It's no different with my machine teammate. Perhaps it provides me an answer and I can act on that, but if I want to investigate further, I want a way to be able to

    access that information and that reasoning. And if I can't, that may impact my willingness, my trust in that teammate (human or machine).

19. Will the scenario and accompanying documentation be released so vendors that aren't selected are able to understand the requirements to improve offerings in the future?
    a. We will continue to release TMs models and post future RFI’s on sam.gov. Actual data out of the DASH will not be released to non-selected Vendors to ensure IP protection.
20. Can we participate in multiple DASHES?
    a. Yes.
21. Is Space domain of interest for sensing?
    a. Yes, all domains are of interest for sensing.
22. Can you expand on our individual access to the crew/operators. Will we be able to solicit subjective feedback before/during/after scenarios and demonstrations?
    a. Yes. The whole intent of the event is that you observe and interact with the Battle Performers participating in the scenario. Additionally, they will be available to answer questions and provided feedback during your development portions of the schedule. We want you to have a rich, robust interaction with the Battle Performers and garners as much data as you can to conduct your development.
23. Should humans be confirming the decisions / effects, or does the machine make decisions, choose effects, and the humans can review them later?
    a. The way we envision this happening is the microservice will answer the question that PAE is responsible for answering, which is are there any plausible actions that I could take with respect to a particular entity of interest. There will be many such entities of interest. There's a lot more battle management that follows from there. We want you focused on this one micro service and then the rest would be up to the crews and how we're handling this in this scenario. For instance, a crew member may read something on chat that they realize this is something that applies to an objective that's not in my AO but is in another AO. The machine can do the same thing because it knows and has access to those objectives. It could process PAE and then offer up to the crew to say this goes to your neighboring AO. This experimentation campaign (and the Transformational Model as a whole) focus on Human-Machine Teaming. Any combination of human and machines teaming for a solution is acceptable.
24. What is the path to acquisition if selected and solution knocks it out of the park?
    a. This is not an acquisition-based event and there is no selection requirement. We are part of the requirements community and not

acquisition community. This ABMS CFT's Decision Advantage Human-Machine Team events is part of our series of engagement with Industry partners at the experiment level.

25. Are there multiple "Crews" (i.e. 1 for each Development Team) or is there only 1 Crew that will use each of the microservice solutions.
    a. The Battle Performers are assigned to the scenarios and not to any specific service or development team. The crew may swap between scenarios. We'll be running at least 5 scenarios to account for the 5 vendors/development teams to have rich, robust access to the Battle Performers during the events.
26. How many DASH events will there be (per calendar year, and/or overall)?
    a. DASH events are currently planned to occur roughly every 2-3 months.
27. Could non-selected vendors join as observers?
    a. No.
28. Will event & relationship Entities be part of the scenario exercise, or will be constrained to physical entities for this first interaction?
    a. Objects, events, and relationships will all be in play for the scenario.
29. Would there be a time prior to playing out the scenario that documents could be consumed and processed?
    a. Yes.
30. Is the microservice intended to run on each operator tower?
    a. The microservice needs to be available to all Battle Performers. There is no constrained on how to make this happen.

31. Will Cloud based solutions be able to participate?
    a. TBD.
    b. Possible answers
        i. Cloud based solution may or may not be able to interact with the Simulation. If yes, then how and if not, then why not?
32. Can DIS standard be provided? DIS is an IEEE standard. Which version of DIS? The ShOC-N uses DIS version 26.28.
33. For clarity "towers" are desktops. Does H2O have any small servers or similar?
    a. The Tower PCs in use for this DASH event: CPU: Intel Core i7-12700, 2.10 GHz, 32GBs of RAM, GPU NVIDIA RTX A2000
    b. Server: Servers are still TBD. The ShOC-N team is working on an MSCT server that may be available.
34. Will there be 100s of millions of texts estimated to be coming in?
    a. No, but possibly hundreds.
35. Can the interaction with the provided microservice be clarified?

a. The following are possible interactions that the human-machine team (human-operator plus baseline “scope” plus microservice), but this list is not exhaustive:
    i. Microservice offers and recommends setting-adjustments for its own PAE logic given current battle conditions...human accepts or adjusts and applies settings.
    ii. Microservice detects* a new EntityOfInterest, runs PAE, publishes stream of BattleEffects in accordance with human-preselected settings, and presents summary with opportunity for human to view results and veto publishing.
    iii. Microservice detects a new EntityOfInterest, runs PAE, presents the resulting stream of BattleEffects with TimeWindows to human for validation and publication. Human publishes BattleEffects to other decision makers or reasons over presented BattleEffects (e.g., to build a BattleCOA).
    iv. Human detects new EntityOfInterest, inputs EntityOfInterest into Microservice interface. Microservice runs PAE, publishes stream of BattleEffects in accordance with human-preselected settings.
    v. Microservice detects a new EntityOfInterest, runs PAE, finds a TransformationObjective conflict, and brings conflict to human’s attention.
    vi. Microservice offers various ways to query the streamed outputs of a PAE instance. Human formulates query with available Microservice controls, and reasons over query outputs presented by Microservice.
    vii. Human detects new/changed TransformationObjective, inputs new information into Microservice interface and requests Microservice to recalculate previous PAE outputs. Microservice re-runs PAE on all or selected EntitiesOfInterest.

    *For this DASH, every Entity (on the data link & mentioned in chat or voice) will be considered an EntityOfInterest (i.e., your microservice doesn’t have to decide whether an Entity is interesting or not. It should consider all Entities in its awareness as interesting.

36. Will there be a centralized location and broker available to allow data to be disseminated? (like AMQ)
    a. Mission products will be distributed to all participants; simulation data will be brokered via DIS and data link processor.